The Role of Data Science in Space Mission Success

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15th of April 2026

Link to GitHub Project

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- This project aimed to predict the success of Falcon 9 first stage landings, a critical contributor to SpaceX's cost efficiency through rocket reusability.

- A number of data sources were available containing historical information on the landing success rate of the Falcon 9 first stage. Data collected from the SpaceX API and web-based sources such as Wikipedia was cleaned, transformed, and converted into DataFrames for further analysis and for building a machine learning model to predict the expected outcome.

- Using the DataFrames generated with Python's pandas library, several key insights were identified during the Exploratory Data Analysis (EDA) phase of the project. Relationships were observed between multiple variables, such as the booster version used, payload mass, orbit flown, and the corresponding landing and launch sites.

- Given the data analysis results and the defined business requirements, a classification machine-learning approach was deemed suitable for the prediction task. Models including Logistic Regression, SVM, KNN, Random Forest, and XGBoost were implemented and evaluated.

- The model predictions predominantly indicated successful first-stage landings.

Introduction

SpaceX Falcon 9 Overview

The Falcon 9 is a medium-lift launch vehicle developed and manufactured by SpaceX, an American aerospace company founded by Elon Mask. Standing approximately 230 fleet tall with a diameter of 12 feet. It is powered by nine Merlin engines on the first stage and a single Merlin engine on the second stage.

Falcon 9 operates on a two-stage system:

- First Stage (Booster): Provides initial thrust to lift the rocket off the ground. It can return to a landing pad or drone ship for reuse
- Second Stage: Ignites after stage separation to deliver payloads to their target orbit.

Falcon 9 is notable as the first orbital-class reusable rocket. This reusability has dramatically reduced launch costs, making space access more affordable for commercial and government customers.

Project Statement: With the introduction of the Falcon 9 rocket, is it possible to accurately predict whether the first stage will land successfully in future missions?

Data Science Methodology

1 . Business Understanding

- Can Falcon 9 first-stage landing success be predicted accurately?
- Rocket reusability directly impacts launch cost and competitive advantage
- This translates into a data science problem (Binary Classification)

2. Analytic Approach

- Supervised classification task
- Justified the use of classification algorithms due to the binary target variable (success vs failure)
- Identified suitable machine-learning models for prediction

3. Data Requirements

- Identification of required data and features (Historical launch data)
- Identification of the target variable representing landing success

Data Science Methodology

4 . Data Collection

- Identification of data sources (SpaceX API & Wikipedia)
- Data retrieved programmatically and stored for processing

5. Data Understanding & Preparation

- Data cleaning, transformation and feature engineering (Pandas DataFrame)
- Prepared datasets for analysis and modeling

6. Exploratory Data Analysis (EDA)

- Insight discovery and visualization
- Explored relationships between landing success and other features

Data Science Methodology

7. Modeling

- Model selection, training and comparison
- Training multiple classification models
- Applied consistent training and testing process through the use of a ML pipeline

8. Model Evaluation

- Model performance metrics and interpretation
- Evaluating models using various metrics
- Compare results across models

9. Deployment/Conclusion

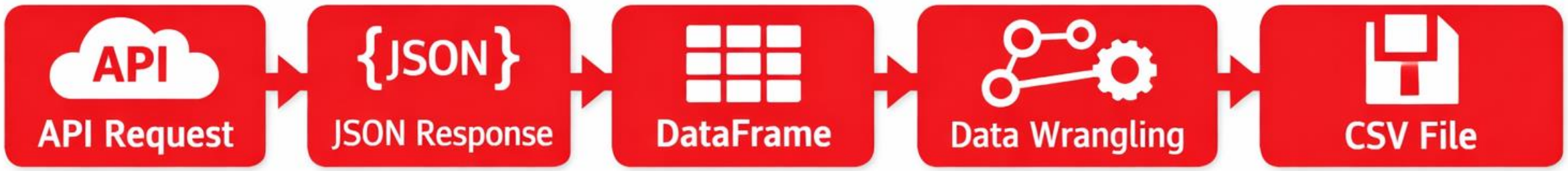
- Interpretation of results and business insight
- Summarize findings and limitations

Methodology – Data Collection

Data Collection Process followed:

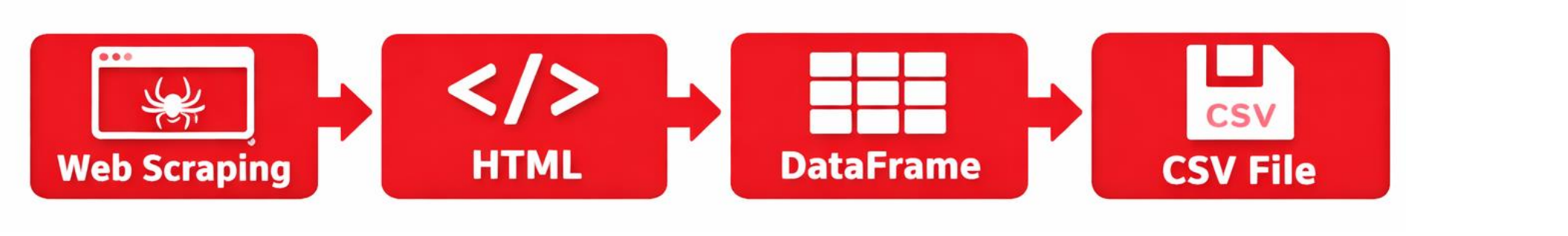
- Collected data from:
  - SpaceX API
  - Wikipedia
- Data inspected
- Handling of missing values
- Create target variable (Column)
- Transform to DataFrame
- Save to CSV file

Methodology – SpaceX API



- Purpose: Build dataset for predictive modelling
- Data Source: SpaceX API - [SpaceX-API/docs at master · r-spacex/SpaceX-API · GitHub]
- Request process: Python Request Library – GET request
- Data File: JSON (Dictionary)
- Convert to Pandas DataFrame
- Data Wrangling: Handing of missing data (Imputation)
- Save to CSV file
- Resulting dataset – See appendix

Methodology – Web Scraping



- Purpose: Build dataset for predictive modelling - Additional information HTML table content
- Data Source: - Falcon 9 Launches (Wikipedia)
- Request process: Web Scraping using Python's BeautifulSoup library
- Data File: HTML – Extract column names from a table
- Convert to Pandas DataFrame
- Save to CSV file
- Resulting dataset – See appendix

Data Wrangling (Feature Engineering)

- Purpose: Prepare DataFrame dataset for Predictive Modelling
- Data Source: - CSV file (Pandas Library)
- Perform EDA
- Create “Class” column – Target Column (Class = 1 if success landing, else 0)
- Save to new CSV file
- Resulting dataset – See appendix

Methodology – Exploratory Data Analysis

During the EDA process the following visualization charts were created using Python's Seaborn and Matplotlib libraries to explore insight of relationships between different variables:

- Scatter Plot – Payload_Mass vs Flight Number (Success of payload size over time)
- Scatter Plot – Launch Site vs Flight Number (Different launch site success over time)
- Scatter Plot – Payload_Mass vs Launch Site (Payload success across different launch sites)
- Bar Graph – Success Rate vs Orbit Type (Orbit type with the highest success rate)
- Scatter Plot – Flight Number vs Orbit Type (Success of landings compared with orbit types)
- Scatter Plot – Payload_Mass vs Orbit Type (Compare orbit success rate with heavier play loads)
- Line Graph – Launch Success Rate vs Yearly Trend (Improving landing success rate over time)

Methodology – Exploratory Data Analysis

Using SQLite, data from the dataset was queried to gain further insight into potential variables that could form part of the predictive modeling process:

- Names of unique launch sites
- 5 records where launch site names started with “KSC”
- Total payload carried by boosters from NASA
- Average payload mass carried by booster version F9 v1.1
- Date of the first successful landing outcome on the drone ship
- Names of boosters successfully landed on a drone ship with a payload mass between 4000 kg and 6000 kg
- Total number of successful and failed mission outcomes
- Names of the boosters carried the maximum payload mass
- For 2017 , months booster versions and launch site landing outcomes were successful
- Counting of landing outcomes (Success or Failed) for the drone ship and ground pad for the period 04/06/2010 to 20/03/207

Methodology – Exploratory Data Analysis

Using the Folium library, map objects were created to plot the locations data for each launch site:

- A red circle and a marker for each launch site – Visually mark the exact geographical location of each launch site
- Coloured labels on each launch site – Red label indicating failed launches and green label indicating successful launches
- Mouse position coordinates – Determine the coordinates of the coastline, railway, highway and city close to launch site CCAFS CL -40
- Blue line and a marker with the distance to the closest coastline, railway, highway and city to CCAFS LC-40

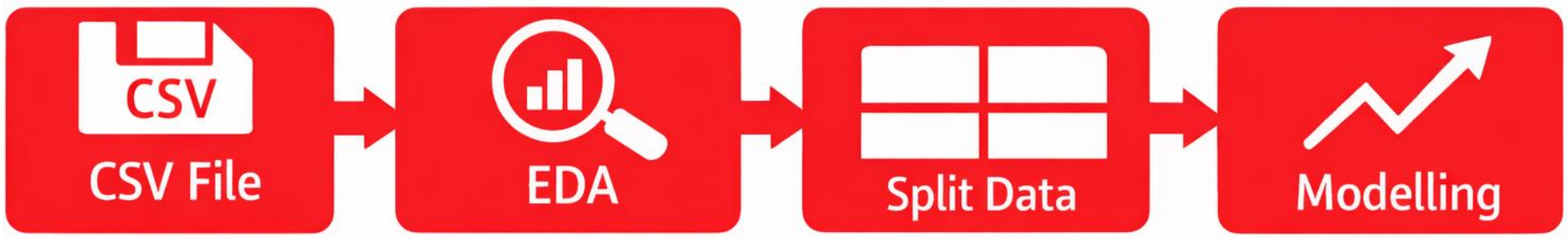
Methodology – Exploratory Data Analysis

Using the Plotly Dash library, an interactive dashboard was created to gain further insight to the different launch sites, payload data and booster versions used. Available from the dashboard:

- A pie chart with an interactive filter to display the success rate for all launch sites, or when individual launch sites are selected the pie chart will display the Success vs Failed launch percentage for the site

- A bar graph with an interactive horizontal slider to display the Launch Outcome Count (Success vs Failure) for each Booter Version over a selected Payload (kg) range.

Modelling – Predictive Analysis (Train & Test)



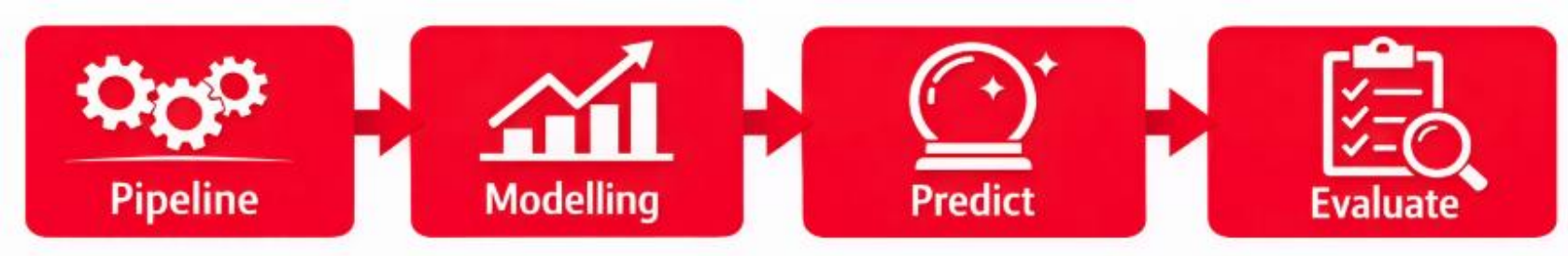
Moving to the Modelling step of the data science methodology, predictive analysis must be performed on the available cleaned datasets to:

- Determine (EDA) and extract the required variables (including target value) from the datasets to be used in the modelling process

- Split the extracted data into a train and test set

- The following classification models were tested and results evaluated:
  - Logistic Regression with a GridSearchCV object
  - SVM (Support Vector Machine) with a GridSearchCV object
  - Decision Tree classifier with a GridSearchCV object
  - KNN (Known Nearest Neighbors) classifier with a GridSearchCV object

Modelling – Predictive Analysis (Pipeline)



In addition to the normal flow of data modeling a ML pipeline was used to:

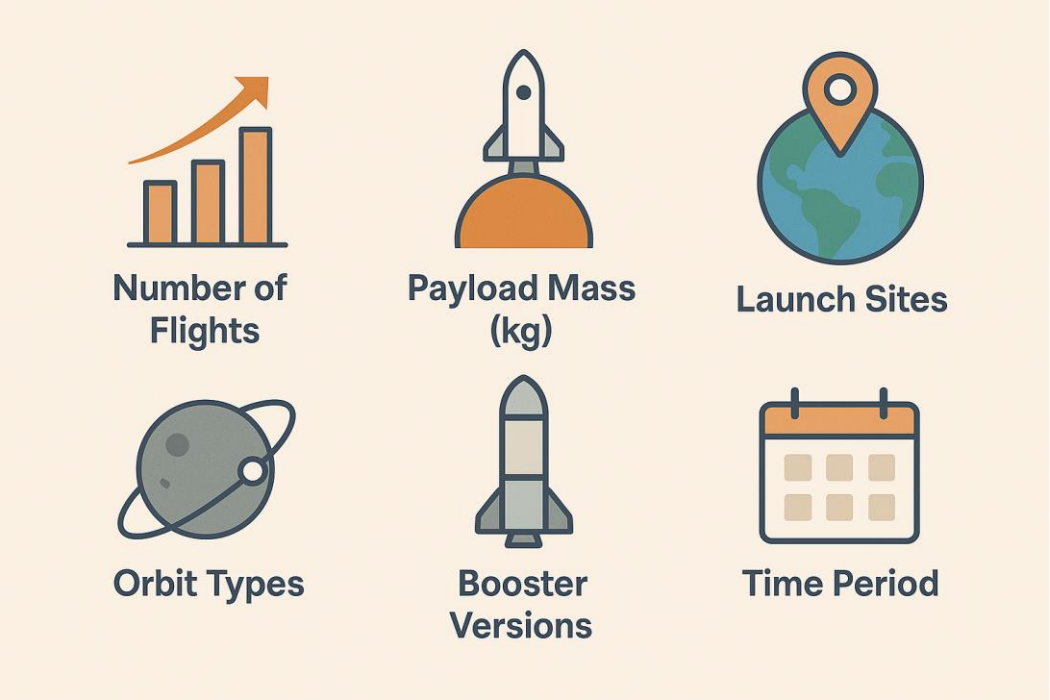
Automate and organize the entire modeling process into a single, reproductive flow. It ensures that every step – from data preprocessing to prediction and evaluation – happens in the correct order and under consistent conditions. A pipeline also has the ability to easily plug in any model algorithm to be used in the model.

- For the pipeline model flow Random Forest and XGBoost were tested and evaluated
- Cross-validation was included in the pipeline to assist with the tuning of hyperparameters across all the steps
- Reference: https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%2010/10.2/SpaceX_Machine%20Learning%20Prediction_pipeline.ipynb

Results - Exploratory Data Analysis

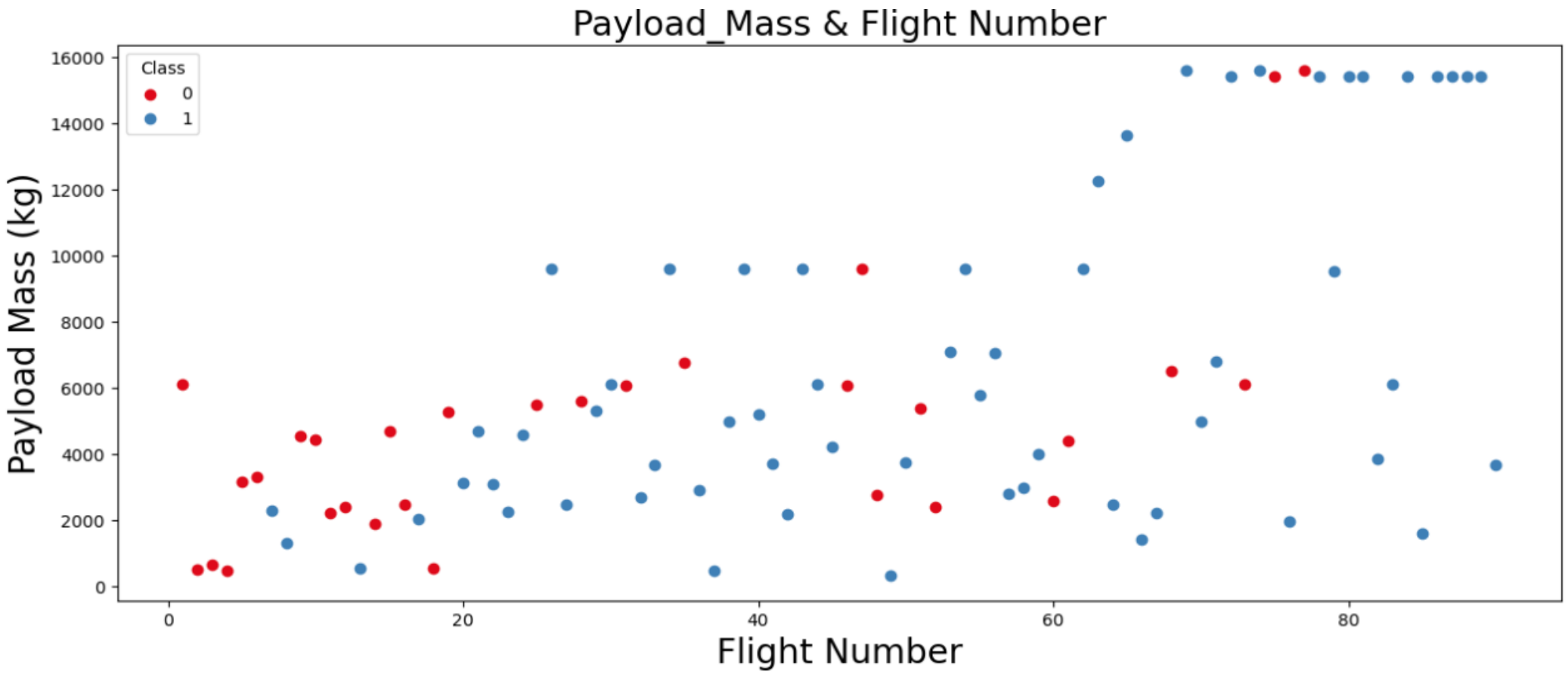
From the prepared datasets the following data variables were used to perform Exploratory Data Analysis to determine the relationships between the "Variables" and the "Success Rate" of the launches:

- Number of Flights
- Payload_Mass (kg)
- Launch Sites
- Orbit Types
- Booter Versions
- Time Period



Results - Exploratory Data Analysis (Visualization)

- General trend: As the flight number increases, the payload mass tends to rise. Overtime, SpaceX has been launching heavier payloads
- Red dots (Class 0) – Booster not landing successfully
- Blue dots (Class 1) – Booster land successfully



Results - Exploratory Data Analysis (Visualization)

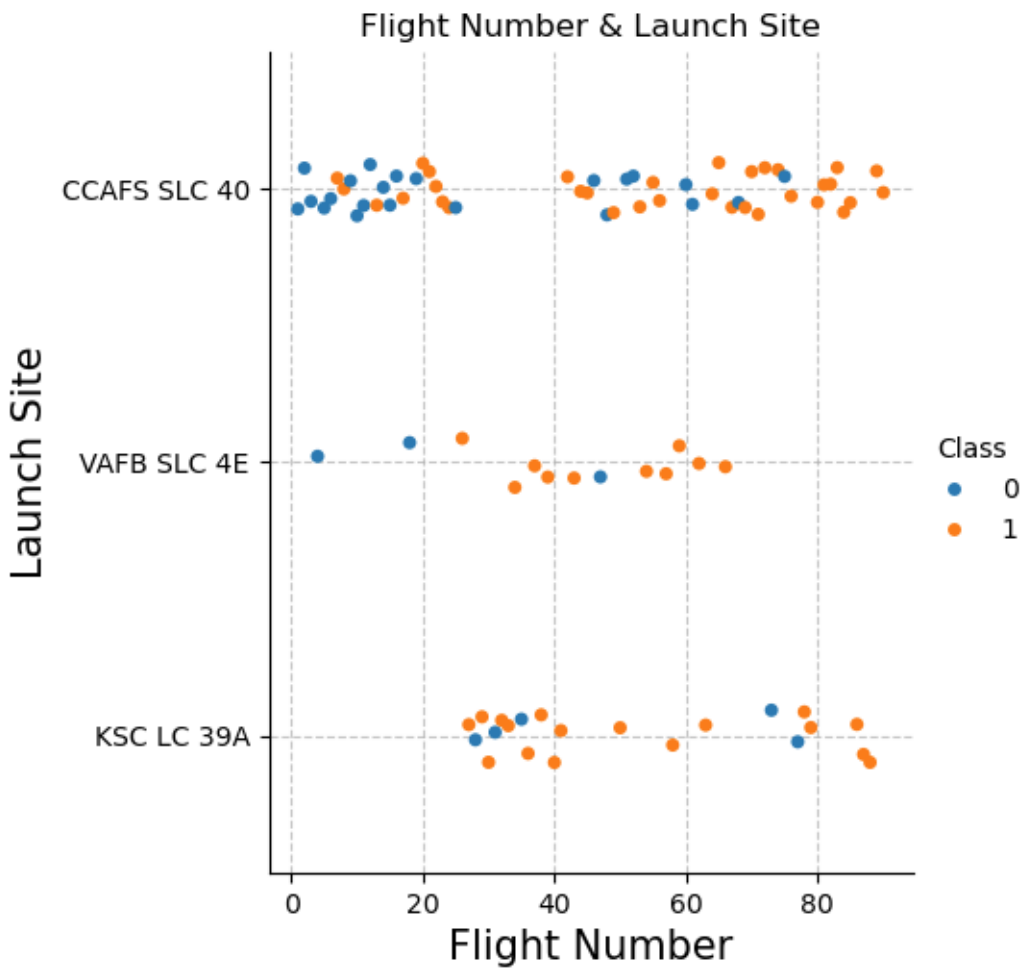
- Scatter plot “Flight Number & Launch Site” gives a clear picture of how SpaceX’s launch success varies across different launch sites and over time.

- Colour Coding:
  - Blue (Class 0) – Did not land successfully
  - Orange (Class 1) – Landed successfully

- As flight numbers increase, Class 1 outcomes become dominant, particularly at KSC LC 39A, suggesting improved reliability and technology

- Outcome demonstrates progressive improvement in landing success across SpaceX’s launch history.

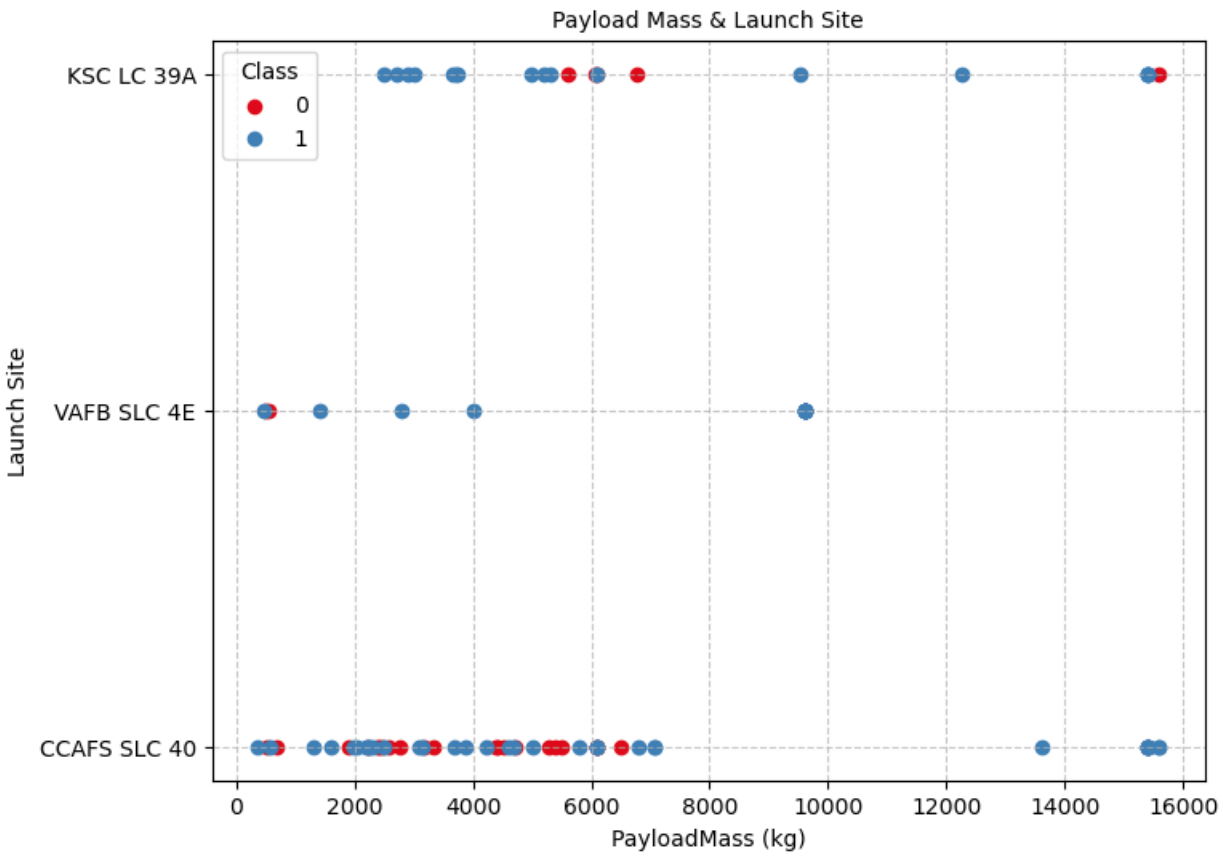
- Kennedy Space Center (LC39A) emerges as the most consistent site for successful booster recovery



Results - Exploratory Data Analysis (Visualization)

- Scatter plot “Payload Mass & Launch Site” highlights how payload size and launch site relate to booster landing success

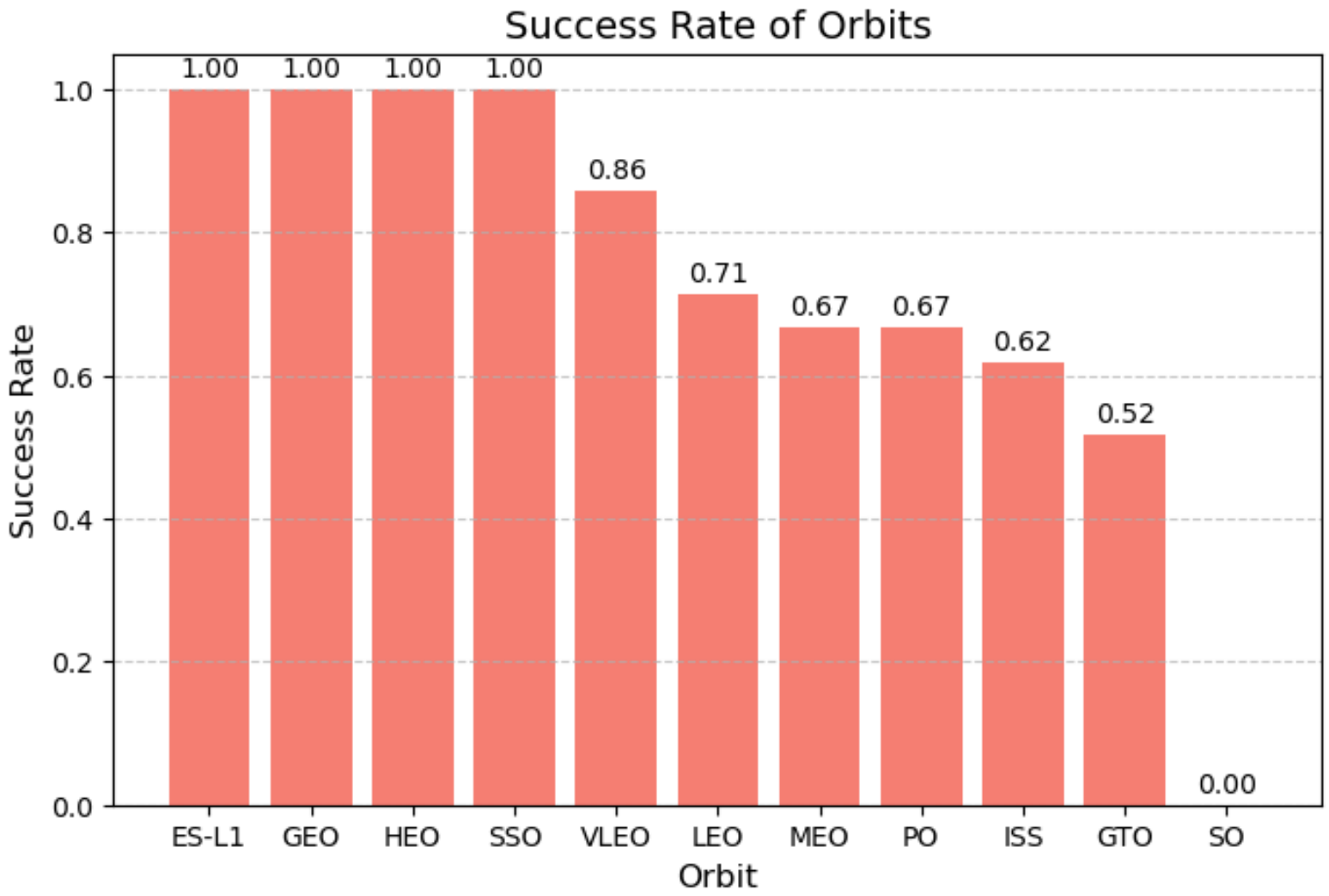
- The outcome demonstrates that payload capacity has increased significantly over time, and successful landings are now achieved across all launch sites, even with heavy payloads



Results - Exploratory Data Analysis (Visualization)

- The bar chart “Success Rate of Orbits” shows how booster landing success varies depending on the orbital destination

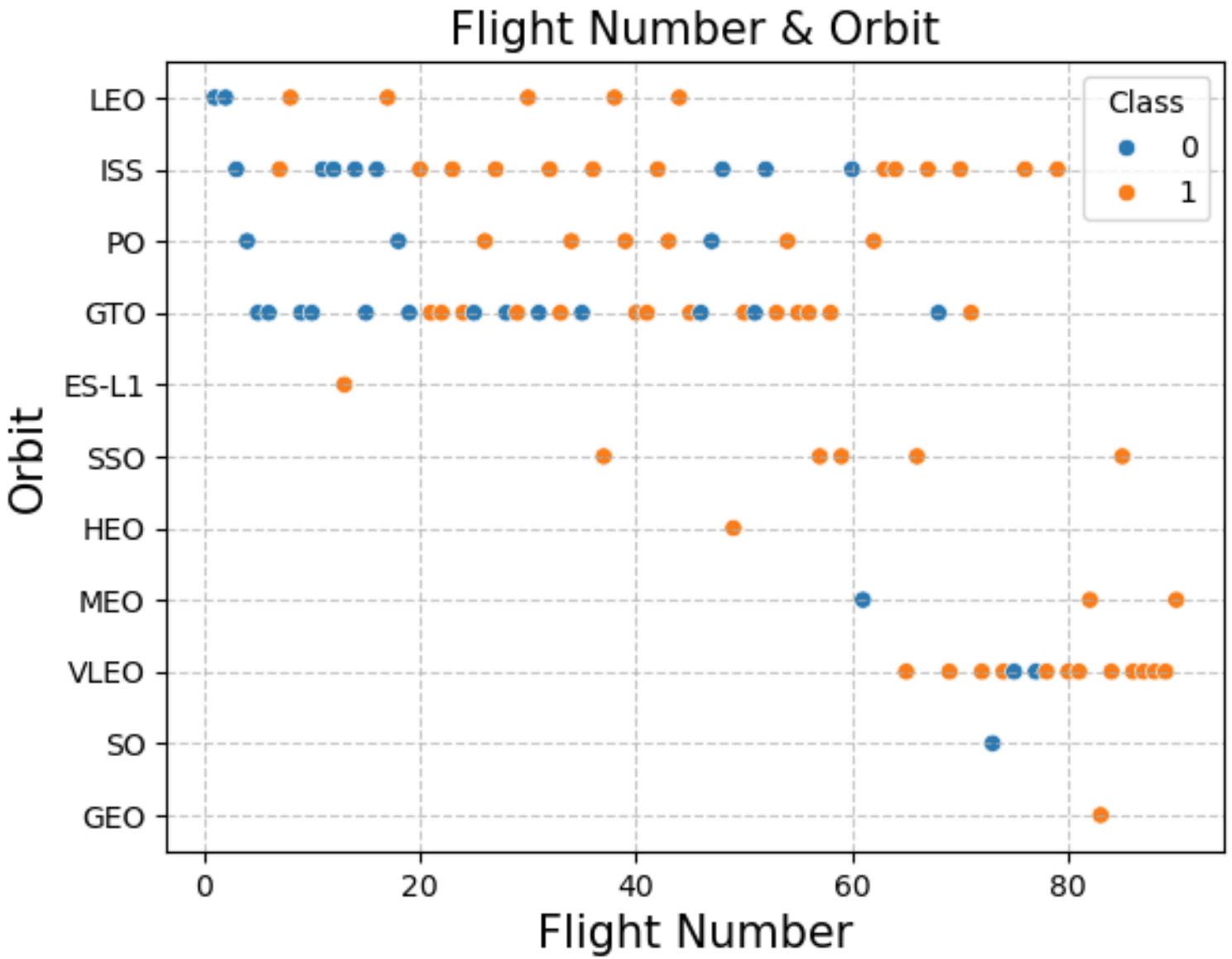
- The outcome demonstrates that success rates are orbit-dependent.



Results - Exploratory Data Analysis (Visualization)

- The scatter plot "Flight Number & Orbit" shows how SpaceX missions are distributed across different orbital destinations and how booster landing success has evolved over time

- Class Outcomes:
  - Orange (Class 1) -> Successful booster landings
  - Blue (Class 0) -> Unsuccessful landings

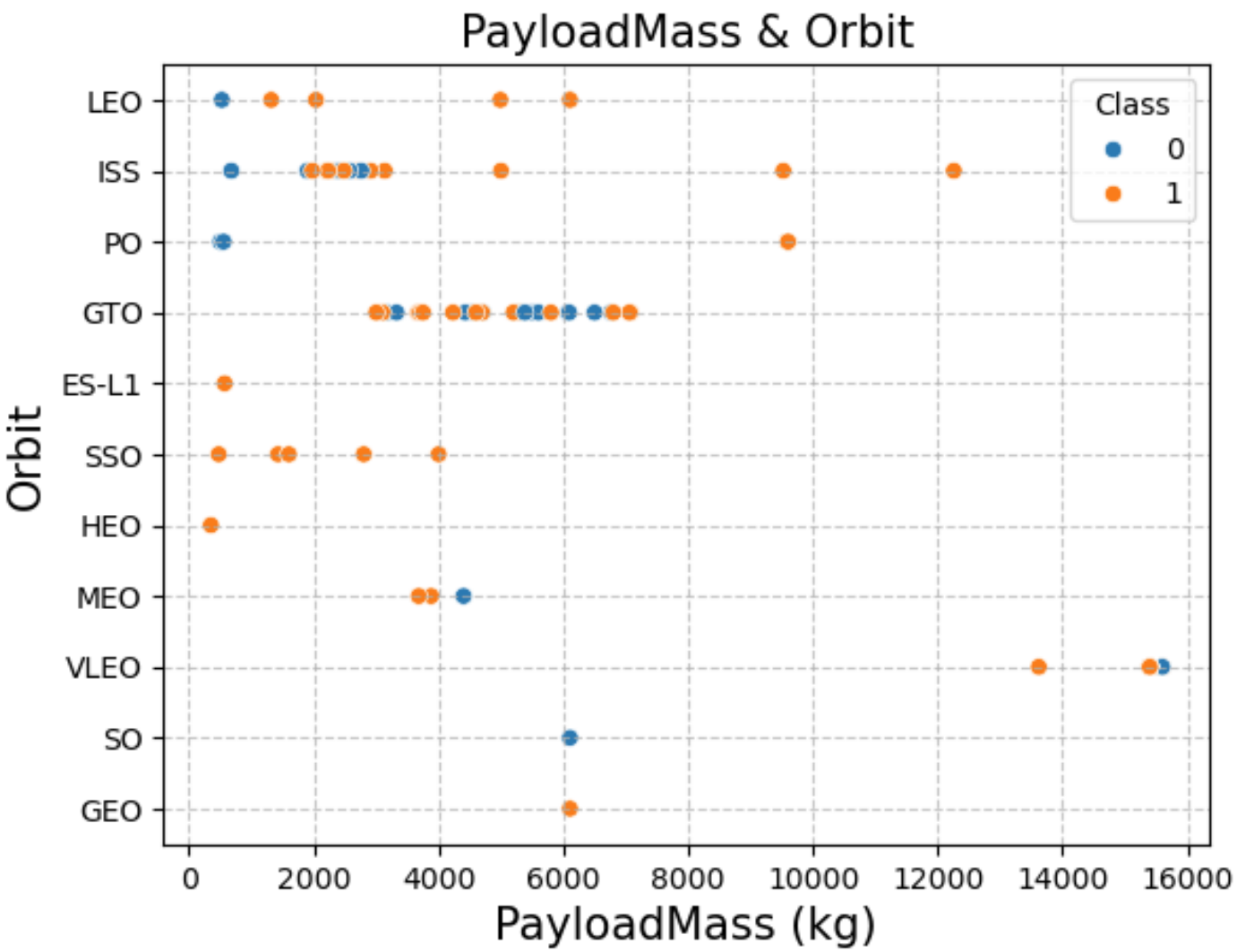


Results - Exploratory Data Analysis (Visualization)

- The scatter plot "PayloadMass & Orbit" shows how payload size related to orbital destination and booster landing success

- Class Outcomes:
  - Orange (Class 1) -> Successful booster landings
  - Blue (Class 0) -> Unsuccessful landings

- The plot demonstrates that payload mass alone does not determine success

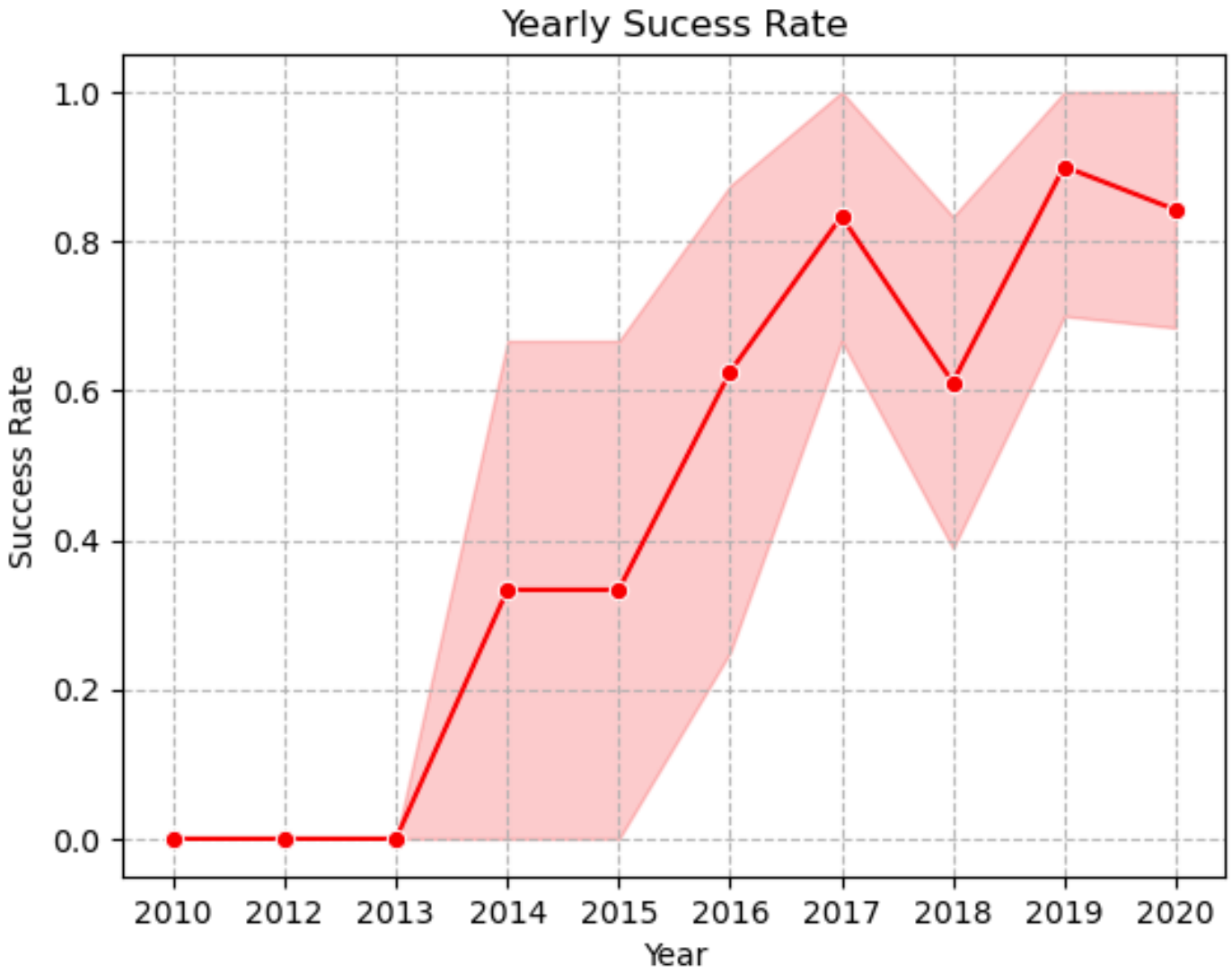


Results - Exploratory Data Analysis (Visualization)

- The line graph " Yearly Success Rate" shows how booster landing success evolved between 2010 and 2020

- The outcome highlights rapid technological progress: from zero success in the early 2010s to consistently high success rates by the late 2017s.

- The dip in 2018 shows that setbacks occurred, but the overall trajectory is upwards, reflecting SpaceX mastery of booster recovery over the decade



Results - Exploratory Data Analysis (SQL)

Launch Sites extracted from the prepared dataset:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site	State (USA)	Coordinates (Latitude, Longitude)
CCAFS LC-40 (Cape Canaveral Space Force Station)	Florida	28.5619° N, −80.5772° W Wikipedia
VAFB SLC-4E (Vandenberg Space Force Base)	California	34.633° N, −120.613° W Wikipedia
KSC LC-39A (Kennedy Space Center, Merritt Island)	Florida	28.6083° N, −80.6044° W Wikipedia

Notes: CCAFS SLC -40, same as LC -40, sometimes written differently

Results - Exploratory Data Analysis (SQL)

Names of the boosters which have success in ground pad and have payload mass between 4000kg and 6000kg

Date	Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
2017-05-01	F9 FT B1032.1	5300	Success (ground pad)
2017-09-07	F9 B4 B1040.1	4990	Success (ground pad)
2018-01-08	F9 B4 B1043.1	5000	Success (ground pad)

Results - Exploratory Data Analysis (SQL)

Total number of Successful and Failure mission outcomes:

Mission_Outcome	num_missions
Success	98
Success (payload status unclear)	1
Success	1
Failure (in flight)	1

mission_outcome	num_missions
Success	100
Failure	1

Results - Exploratory Data Analysis (SQL)

Records displaying the month names, successful landing_outcomes in ground pad, booster versions and launch site for months in the year 2017:

Year	MonthName	Landing_Outcome	Booster_Version	Launch_Site
2017	02	Success (ground pad)	F9 FT B1031.1	KSC LC-39A
2017	05	Success (ground pad)	F9 FT B1032.1	KSC LC-39A
2017	06	Success (ground pad)	F9 FT B1035.1	KSC LC-39A
2017	08	Success (ground pad)	F9 B4 B1039.1	KSC LC-39A
2017	09	Success (ground pad)	F9 B4 B1040.1	KSC LC-39A
2017	12	Success (ground pad)	F9 FT B1035.2	CCAFS SLC-40

Results - Exploratory Data Analysis (SQL)

Records displaying the month names, successful landing_outcomes in ground pad, booster versions and launch site for months in the year 2017:

Year	MonthName	Landing_Outcome	Booster_Version	Launch_Site
2017	February	Success (ground pad)	F9 FT B1031.1	KSC LC-39A
2017	May	Success (ground pad)	F9 FT B1032.1	KSC LC-39A
2017	June	Success (ground pad)	F9 FT B1035.1	KSC LC-39A
2017	August	Success (ground pad)	F9 B4 B1039.1	KSC LC-39A
2017	September	Success (ground pad)	F9 B4 B1040.1	KSC LC-39A
2017	December	Success (ground pad)	F9 FT B1035.2	CCAFS SLC-40

Results - Exploratory Data Analysis (SQL)

Result: Count of landing outcomes between 04-06-20210 and 20-03-2017

- Drone Ship – Failure
- Ground Pad - Success

Year	MonthName	Landing_Outcome	Booster_Version	Launch_Site
2017	February	Success (ground pad)	F9 FT B1031.1	KSC LC-39A
2017	May	Success (ground pad)	F9 FT B1032.1	KSC LC-39A
2017	June	Success (ground pad)	F9 FT B1035.1	KSC LC-39A
2017	August	Success (ground pad)	F9 B4 B1039.1	KSC LC-39A
2017	September	Success (ground pad)	F9 B4 B1040.1	KSC LC-39A
2017	December	Success (ground pad)	F9 FT B1035.2	CCAFS SLC-40

Results - Exploratory Data Analysis (SQL)

Year of the first successful Landing_Outcome on a Drone Ship

min(Date)	Landing_Outcome
2016-04-08	Success (drone ship)

Results - Exploratory Data Analysis (Interactive)

The following results are from an interactive map created by Python's Folium package:

- Plotting the launch sites on a geographical map of the United States adds a valuable spatial dimension to the analysis of SpaceX booster performance.
- By anchoring each site to its state and coordinates, you can connect location-specific factors to mission outcomes.
- The following results are relevant to the outcome of this project: "Success Rate" of each launch site

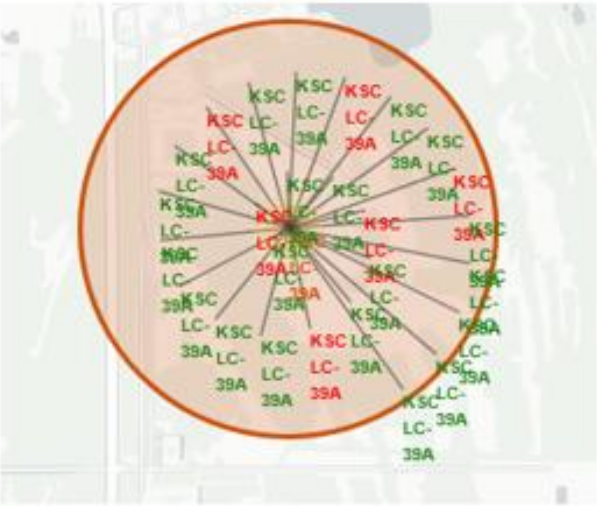
Results - Exploratory Data Analysis (Interactive)

The following results shows the success rate of each launch site:

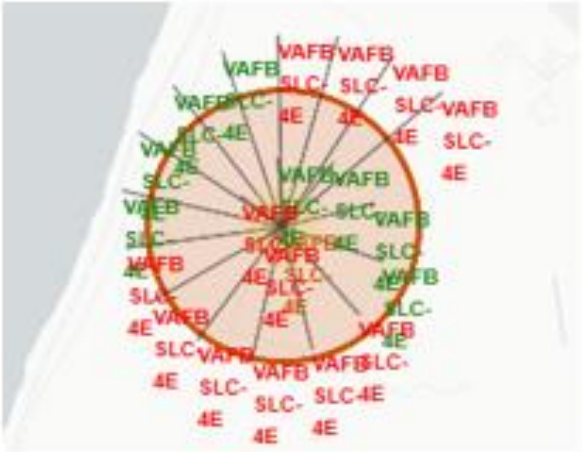
- Coler Coding:
  - Red – Unsuccessful Landings
  - Green – Successful Landings

Result!

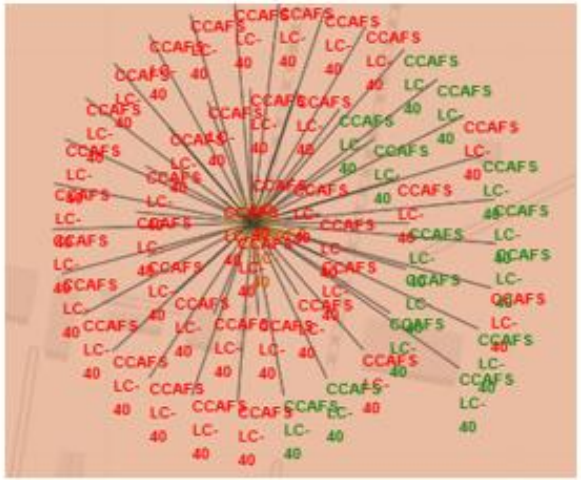
Launch Site: KSC LC – 39A



KSC LC -39A



VAFB SLC -4E



CCAFS LC -40

Results - Exploratory Data Analysis (Interactive)

The following results are from an interactive Plotly Dashboard

To gain deeper insight into the success rate of the SpaceX launches, it's useful to compare three dimensions together:

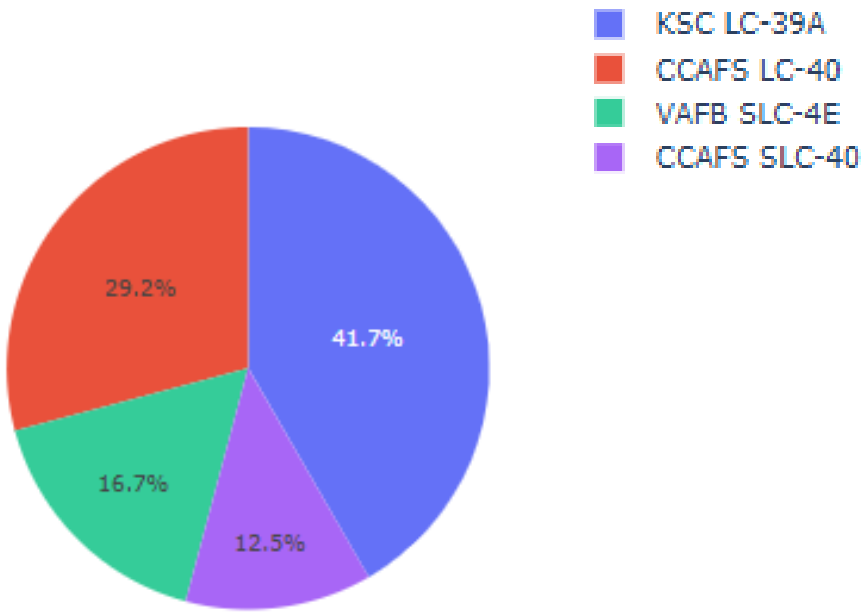
- Launch Site: (CCAFS LC -40, VAFB SLC – 4E & KSC LC -39A)
- Booster Versions: (B4, FT, v1.0, v1.1 & B5)
- Payload Mass Range: 0 to 1000 (Kg)

Results - Exploratory Data Analysis (Interactive)

Results from the Pie Graph shows the combined total Success Launches by Site

SpaceX Launch Records Dashboard

Result!

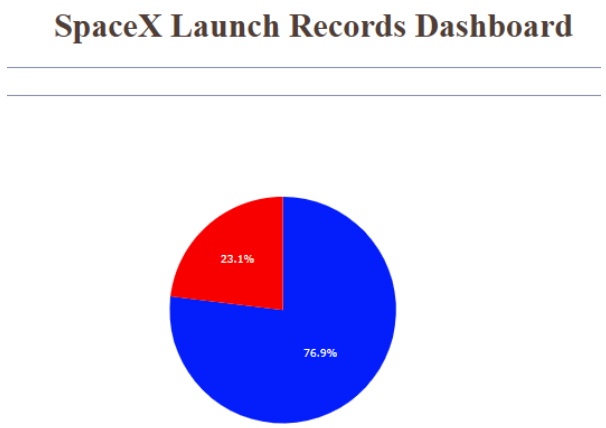


From the pie chart is it event, like with other results presented, launch site KSC LC -39A recorded the most successful launches

This shows Florida sites dominate launch activity, with Kennedy LC -39A leading in successful recoveries

Results - Exploratory Data Analysis (Interactive)

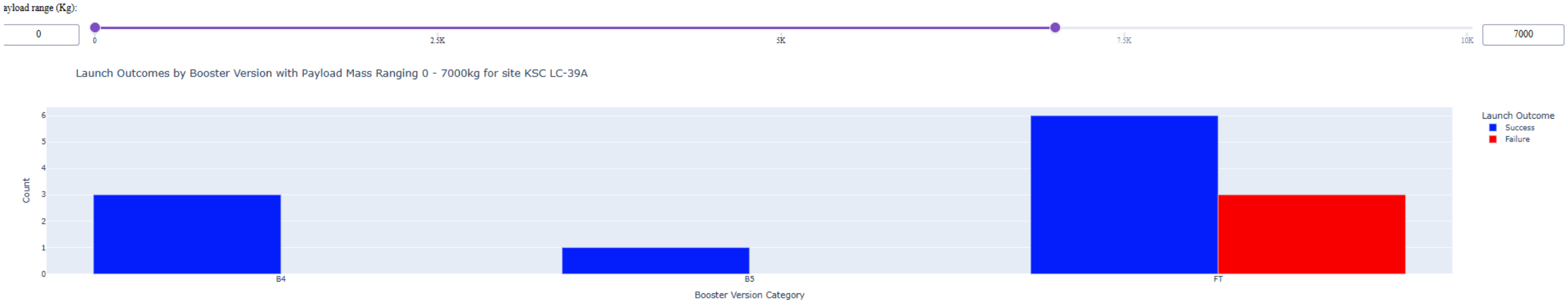
Results from the Bar Chart – Outcomes by Booster Version and corresponding payload range



Insight for Launch Site KSC LC – 39A:

- Early boosters succeed mainly with lighter payloads
- Block 5 and FT boosters succeed across the full payload spectrum, especially in heavier ranges

Result!



Results – Predictive Analysis (Modelling)

With the EDA completed the next step is to perform the modelling process to predict the outcome of the successful landing of the SpaceX first-stage booster.

The datasets analysed is structured for a Supervised Classification Model. The target variable is the launch outcome (class: Success vs Failure) and the predictors are the features like payload mass, orbit type, launch site, and booster version.

Why Classification Fits:

- Binary target – Success (1) vs Failure (0)
- Categorical predictors - Launch site, booster version category, orbit type
- Continuous predictors – Payload mass (kg)
- Goal – Predict whether a future launch will succeed given its site, booster and payload

Results – Predictive Analysis (Modelling)

Models Tested:

- Logistic Regression – Interpretable, good baseline for binary classification
- Support Vector Machines (SVM) – Useful if the dataset is balanced and not too large
- Decision Tree – Handle categorical + continuous features well, capture nonlinear relationships
- XGBoost – Powerful for tabular data, often yields high accuracy

The table below shows a summary result of the accuracy performance of each classifier tested

MODEL	CV SCORE	CLASSIFICATION REPORT SCORE	ROC SCORE
LOGISTIC REGRESSION	0.83	0.83	0.91
SVM	0.83	0.83	0.87
DECISION TREE	0.83	0.83	0.70
KNN	0.83	0.83	0.79
RANDOM FORREST	0.86	0.83	0.83
XGBOOST	0.86	0.83	0.89

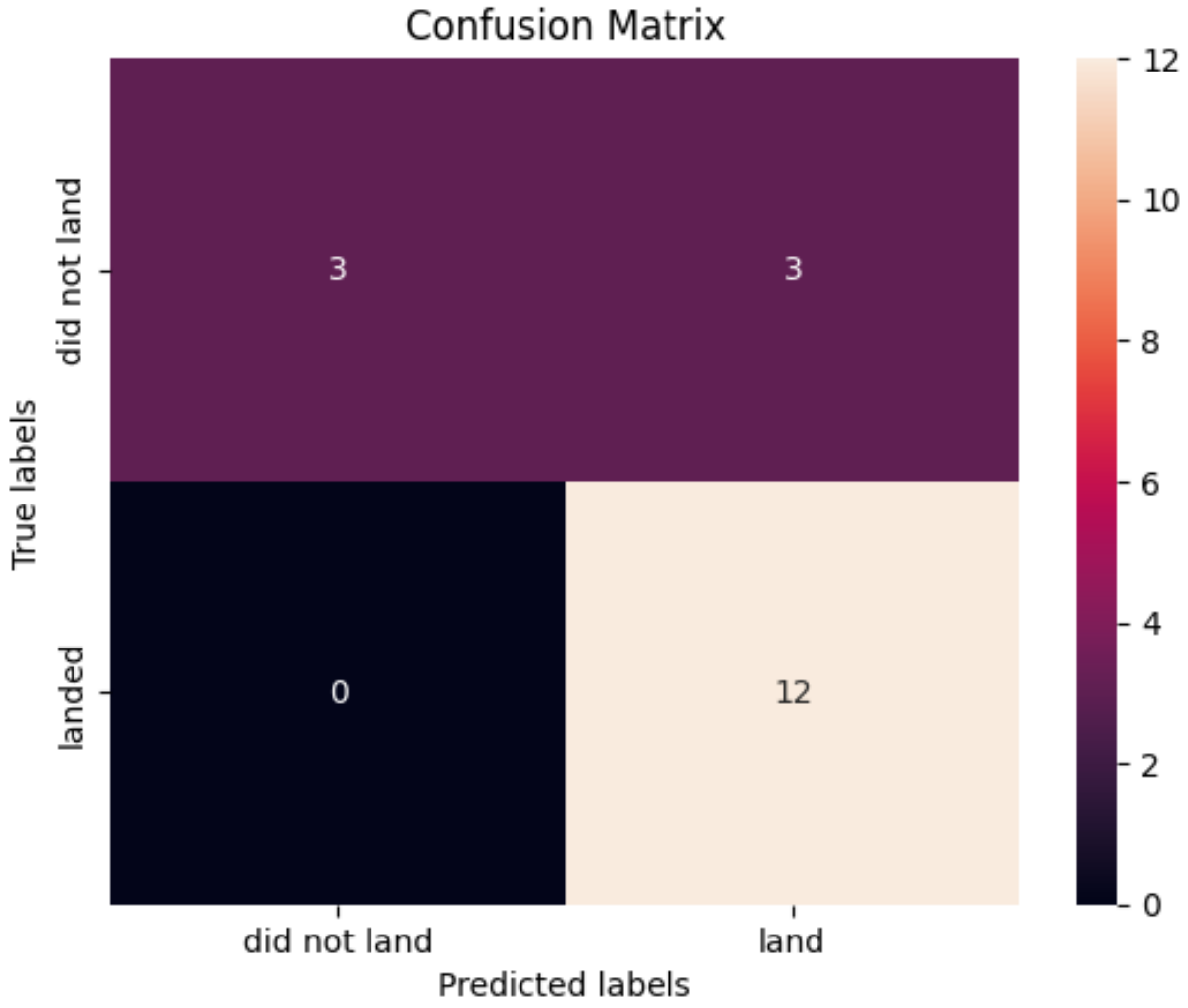
Results – Predictive Analysis (Modelling)

Confusion Matrix Results – How well the classification model predicted the booster landing success vs failure

For all the models tested the Confusion Matrix prediction results are the same

Breakdown of Results:

- True Negatives (3) -> Correctly predicted “did not land”
- False Positives (3) -> Predicted “landed” when the booster actually did not land
- False Negatives (0) -> No cases where the model predicted “did not land” but the booster actually landed
- True Positives (12) -> Correctly predicted “landed”



Results – Predictive Analysis (Pipeline)

For the final model selections a Cross-Validation method was incorporated into a pipeline to:

- Do more reliable performance estimates
- Better generalization check
- Fair model comparison
- Hyperparameter tuning support

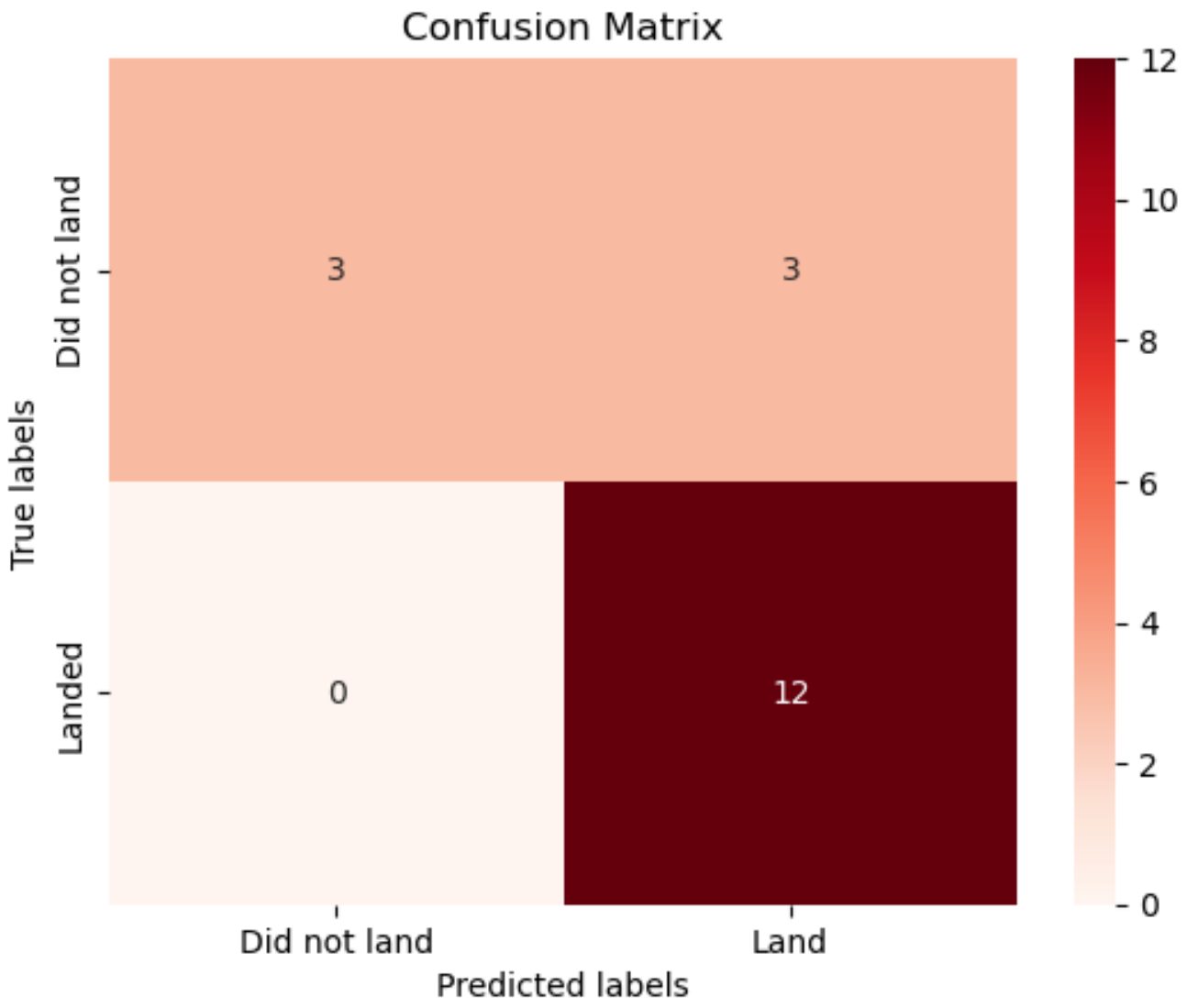
Random Forrest and XGBoost Claissifiers were chosen for the predictive analysis and evaluation

Reference: [Jupyter Notebook Lab - Pipelines]

Results – Predictive Analysis (Pipeline)

Random Forrest (Confusion Matrix Results)

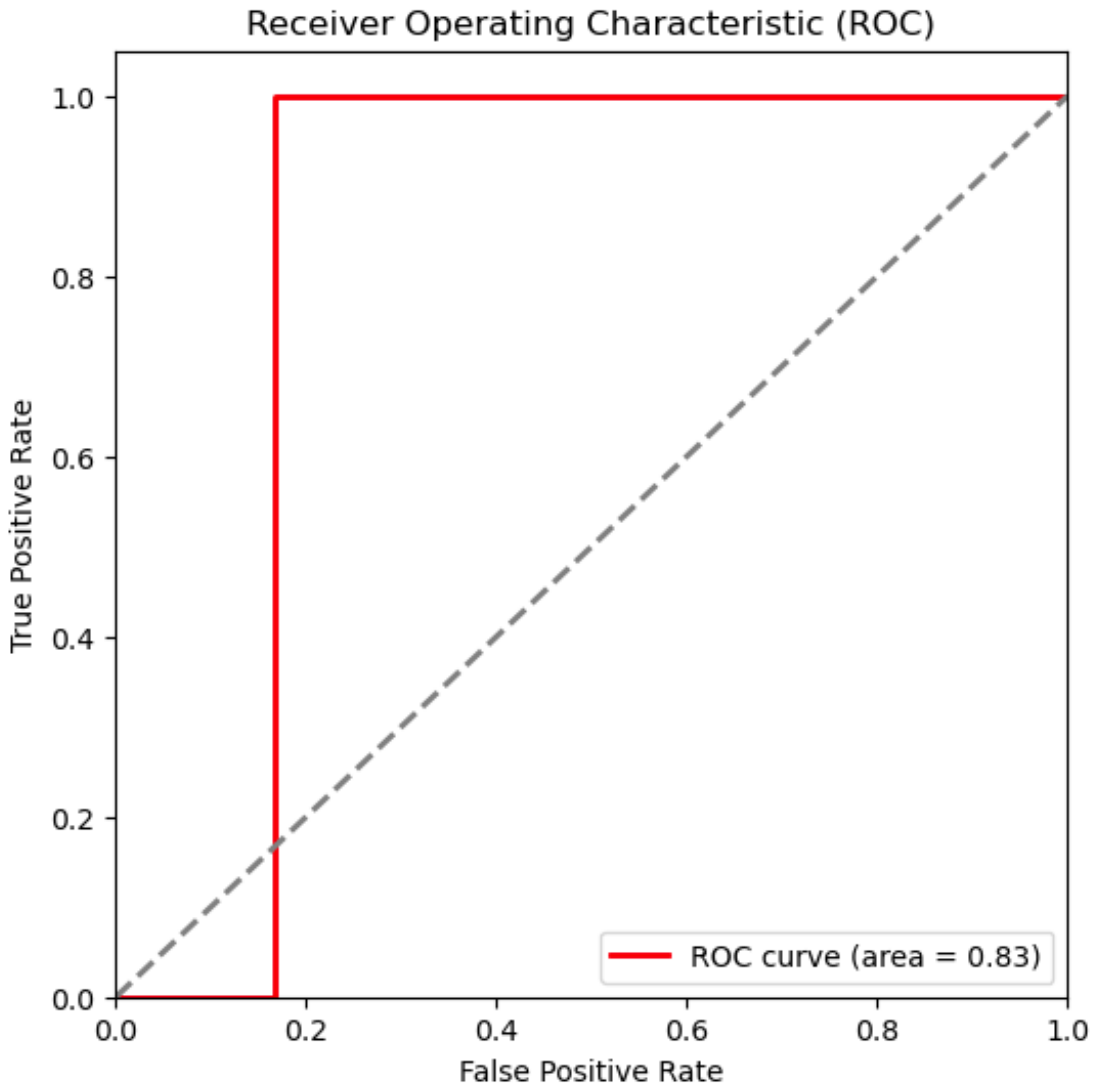
	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18



Results – Predictive Analysis (Pipeline)

Random Forrest (ROC Results)

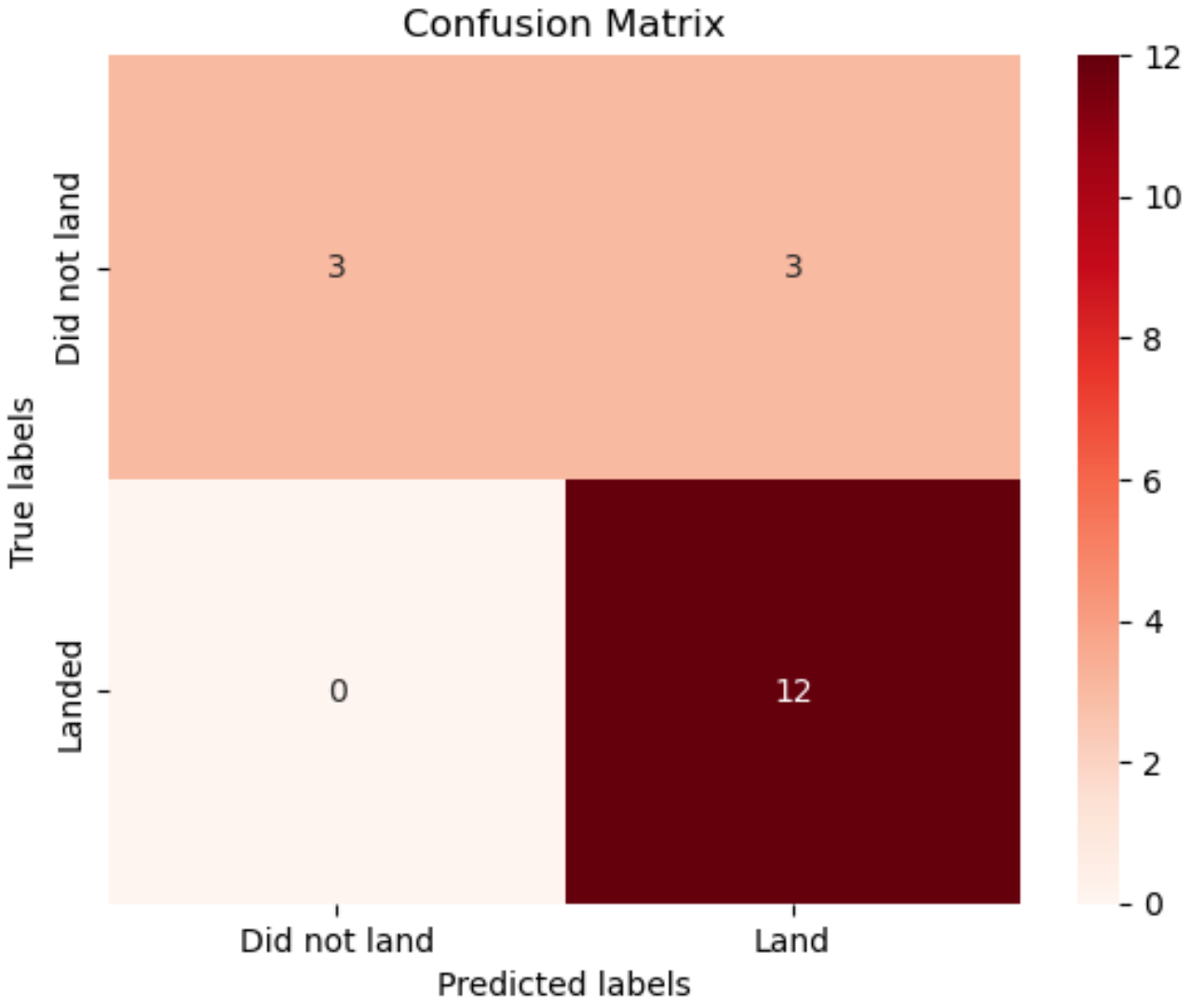
- The ROC curve outcome shows how effectively the classification model distinguished between successful and failed booster landings
- The red ROC curve rises well above the diagonal (gray dashed line), showing that the model achieves a high true-positive rate while keeping false positives relatively low
- The steeper the initial rise, the better the model is at identifying true landings early without many false alarms



Results – Predictive Analysis (Pipeline)

XGBoost (Confusion Matrix Results)

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18



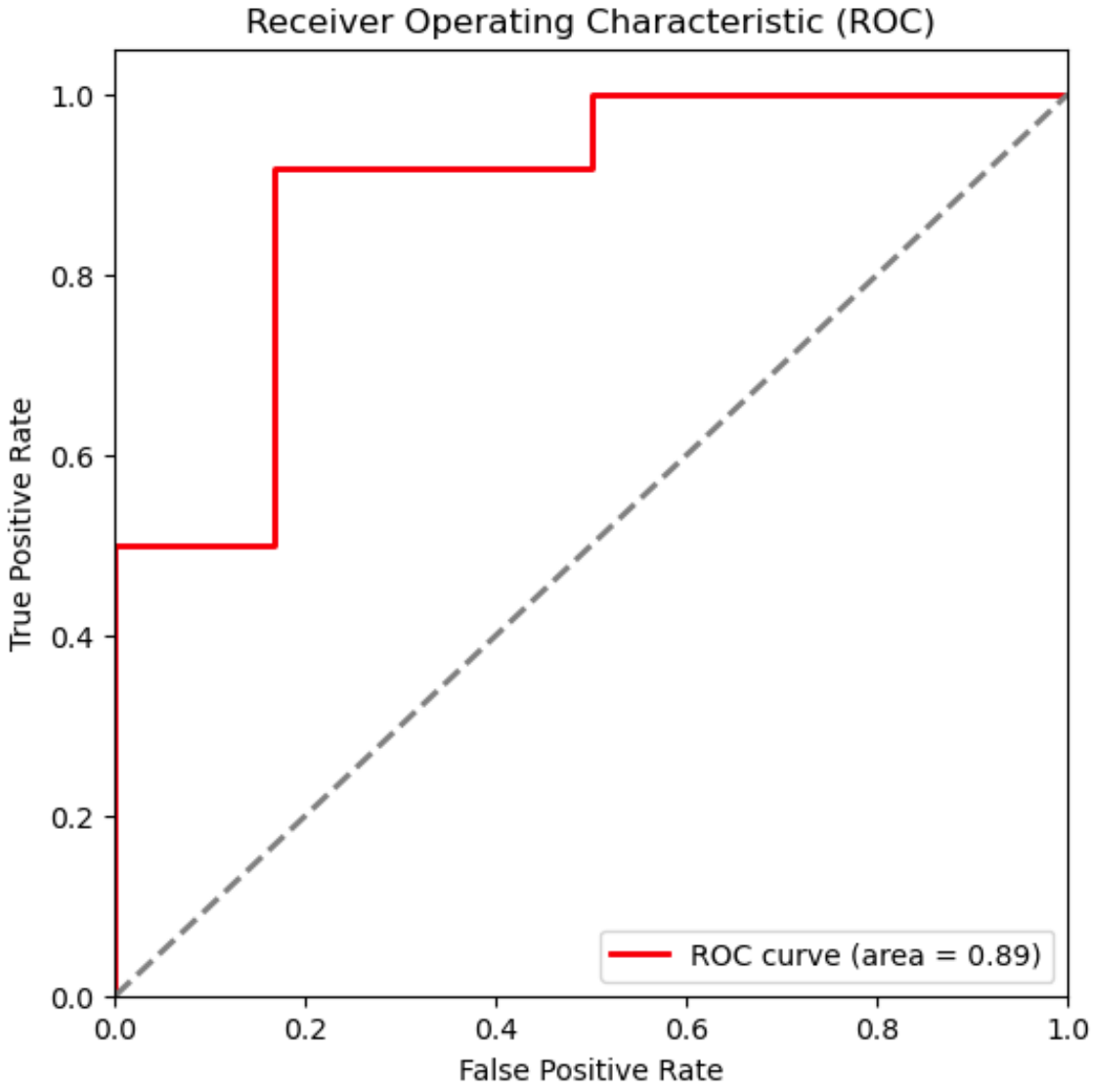
Results – Predictive Analysis (Pipeline)

XGBoost (ROC Results)

- The ROC curve with an AUC of 0.89 indicates that the classification model performs very well at distinguishing between successful and failed booster landings

- The red ROC curve rises steeply toward the top-left corner, meaning the model achieves a high true-positive rate while keeping false positives low. This reflects excellent sensitivity and specificity balance

- The model is highly reliable for predicting booster success, outperforming the Random Forest with AUC = 0.83



Model Evaluation Summary Report

Cross-Validation (CV):

- Logistic Regression, SVM, Decision Tree and KNN achieved CV scores around 0.83 showing consistent baseline performance.
- Random Forest and XGBoost improved to 0.86, demonstrating stronger generalization across folds.
- Cross-validation confirmed that ensemble methods provide more reliable predictions than single models.

Classification Reports:

- All models produced similar classification report scores (~0.83), indicating balanced precision and recall.
- The differences emerged more clearly in ROC and confusion matrix analysis.

ROC Curves:

- Logistic Regression achieved an AUC of 0.91, showing excellent discriminative ability.
- XGBoost followed closely with 0.89, combining strong predictive power with flexibility.
- Random Forest scored 0.83, still robust but slightly weaker in separating classes.
- Decision Tree and KNN lagged behind with lower AUC values, reflecting limited reliability.

Model Evaluation Summary Report

Confusion Matrix Insights:

- The best models (Logistic Regression, XGBoost) correctly identified nearly all successful landings.
- False negatives were minimal, but some false positives occurred (predicting success when failure happened).
- This means the models are highly sensitive to success detection, though precision can be improved with threshold tuning

Overall Outcome:

- Best balance: Logistic Regression (interpretability + high ROC) and XGBoost (strong CV + ROC)
- Robust ensemble: Random Forest also performed well, though slightly weaker in ROC
- Weaker options: Decision Tree and KNN showed lower discriminative ability

Conclusion:

Cross-Validation validated the consistency of Random Forest and XGBoost, while ROC and confusion matrices highlighted Logistic Regression and XGBoost as the most effective models for predicting booster landing success.

Conclusion

Applying the data science methodology provided a structed and effective approach to addressing the problem of predicting SpaceX Falcon 9 first-stage landing success.

Beginning with a clear business and analytic understanding, the project identified rocket reusability as a critical factor influencing launch costs and competitiveness in the aerospace industry.

Data collection from the SpaceX API and supplementary Wikipedia sources ensured a comprehensive dataset, while systematic data wrangling improved data quality and enabled the creation of a meaningful target variable. It was event with the data collection process that only 91 rows of data was available which will impact predictive modelling outcome.

Exploratory data analysis revealed key relationships between landing success and factors such as payload mass, orbit type, and launch site, offering valuable domain insights. Interactive visualizations further enhanced understanding by highlighting geographic and operational patterns. From the data discovery, it is event that over time, the booster landing success rate improved, and one specific launch site was more prominent to the success of the SpaceX Falcon 9 program.

In the modeling phase, multiple classification algorithms were developed, tuned, and evaluated, demonstrating that machine learning techniques can reasonably predict booster first-stage landing outcomes using historical launch data. All models exhibited a high ability to discriminate between successful and unsuccessful landings, as indicated by strong ROC performance. Among the models tested, the XGBoost classifier consistently outperformed the others, making it the best-performing model for the dataset.

Appendix

The following links contain information regarding the datasets and Jupyter notebooks used for this project:

- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.4/dataset_part_1.csv
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.4/dataset_part_2.csv
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.4/spacex_web_scraped.csv
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.4/jupyter-labs-spacex-data-collection-api.ipynb
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.4/jupyter-labs-webscraping.ipynb

Appendix

The following links contain information regarding the datasets and Jupyter notebooks used for this project:

- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.4/labs-jupyter-spacex-data%20wrangling_jupyterlite.ipynb
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.5/dataset_part_3.csv
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.5/edadataviz.ipynb
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%209/9.5/jupyter-labs-eda-sql-edx_sqllite.ipynb
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%2010/10.1/spacex_launch_dash.csv

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The following links contain information regarding the datasets and Jupyter notebooks used for this project:

- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%2010/10.1/spacex-dash-app-bar-plots.py
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%2010/10.1/lab_jupyter_launch_site_location.ipynb
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%2010/10.2/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb
- https://github.com/whitejt1969-hub/ibm_data_science_jan2026_capstone/blob/main/Module%2010/10.2/SpaceX_Machine%20Learning%20Prediction_pipeline.ipynb

Appendix

The following links contain information regarding the datasets and Jupyter notebooks used for this project:

- https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922

Thank you